BREAD (FINAL) ? INSTALLATION MANUAL (PDF SOURCE)

This markdown file is the source for ‘INSTALLATION_MANUAL.pdf‘. It is also included in the submission zip for convenience.

SYSTEM REQUIREMENTS

? **OS**: any OS that supports Docker (Linux/macOS/Windows)
? **Docker**: 24+ recommended
? **Docker Compose**: Docker Compose plugin (v2)
? **Browser**: Chrome/Firefox (latest)
? **Disk**: enough space for Docker images + volumes (Postgres/MinIO)
? **Ports**
- ‘8080‘ (default): gateway (frontend + API proxy)
- ‘5173‘ (optional): ‘frontend-dev‘ Vite dev server
- ‘15432‘ (optional): Postgres port mapping for local debugging
- ‘15672‘ (optional): RabbitMQ management UI
- ‘9001‘ (optional): MinIO console

INSTALL DOCKER

If you don?t have Docker installed yet:

1) Install Docker for your OS
   - Docker Desktop (recommended on Windows/macOS): follow the official guide: https://docs.docker.com/desktop/
   - Linux: install the Docker Engine + Docker Compose plugin (v2) following the official guide: https://docs.docker.com/engine/ and https://docs.docker.com/compose/

2) Start the Docker service
   - Docker Desktop usually starts automatically after installation.
   - On Linux you may need to start/enable the daemon (exact command depends on your distro).

3) Verify Docker is working
‘‘‘bash
docker --version
docker compose version
‘‘‘

If both commands work, you?re ready to run this project with ‘docker compose up -d‘.

REPOSITORY CONTENTS (WHAT IS INCLUDED)

The submission zip is generated from this repository and includes the files needed to run the system, excluding:

? ‘.git/‘
? ‘.env‘ (real secrets)
? ‘node_modules/‘, ‘.venv/‘, ‘__pycache__/‘, build caches, Docker volumes
? ‘dist/‘ (derived artefacts)

In practice, the zip will include ‘docker-compose.yml‘, ‘backend/‘, ‘frontend/‘, ‘nginx/‘, ‘tests/‘, and documentation files like ‘README.md‘, ‘TESTING.md‘, and this manual.

ENVIRONMENT SETUP

1) Create your local environment file:

‘‘‘bash
cp .env.example .env
‘‘‘

2) Adjust variables in ‘.env‘ if needed.

Required for local Docker run (dev defaults provided in ‘.env.example‘):
- ? ‘TZ‘
- ? ‘DJANGO_DEBUG‘
- ? ‘ALLOWED_HOSTS‘
- ? ‘DJANGO_SECRET_KEY‘
- ? Postgres: ‘PG_DB‘, ‘PG_USER‘, ‘PG_PASSWORD‘
- ? RabbitMQ: ‘MQ_USER‘, ‘MQ_PASS‘
- ? MinIO: ‘S3_ACCESS_KEY‘, ‘S3_SECRET_KEY‘, ‘S3_BUCKET‘
- ? ‘PUBLIC_MEDIA_URL‘
- ? ‘GATEWAY_PORT‘

Optional (can be blank):
- ? Gmail OAuth variables (‘GMAIL_*‘)
- ? DeepSeek/LLM variables (‘DEEPSEEK_*‘)

BUILD & RUN WITH DOCKER COMPOSE

From the project root (same folder as ‘docker-compose.yml‘):

Build images:

‘‘‘bash
docker compose build
‘‘‘

Start the stack:

‘‘‘bash
docker compose up -d
‘‘‘

Check status:

‘‘‘bash
docker compose ps
docker compose logs -f
‘‘‘

Stop:

‘‘‘bash
docker compose down
‘‘‘

To remove data volumes (fresh database / fresh MinIO):

‘‘‘bash
docker compose down -v
‘‘‘

DATABASE SETUP (MIGRATIONS / SEED)

? The backend container runs migrations at startup.
? If you need to run migrations manually:

‘‘‘bash
docker compose exec backend python manage.py migrate
‘‘‘

HOW TO ACCESS THE SYSTEM

? **Main entry (frontend + API gateway)**: ‘http://localhost:8080‘ (or ‘http://localhost:${GATEWAY_PORT}‘)
? **Backend API (via gateway)**: ‘http://localhost:8080/api/...‘

Notes:
? The gateway serves the frontend and proxies ‘/api/‘ and ‘/media/‘ to backend/storage.

DEFAULT/DEMO CREDENTIALS

For marking and local testing:
? Verification code is fixed to **‘1234‘** in dev/test mode (no real email required).
? Teacher registration invite code comes from ‘TEACHER_REGISTRATION_INVITE_CODE‘ in ‘.env‘ (default in ‘.env.example‘: ‘12345678‘).
? Some flows require ‘.edu.au‘ emails (see ‘tests/integration/README.md‘ for examples).

TESTING INSTRUCTIONS

See ‘TESTING.md‘ for:
? What tests exist (frontend unit, backend unit, integration helpers)
? How to run each test suite
? What must be tested manually (auth, profile, courses, posts, dashboards, uploads, notifications)

Quick commands:

Frontend unit:

‘‘‘bash
cd tests/frontend-unit
npm install

npm test
‘‘‘

Backend unit:

‘‘‘bash
python -m venv .venv
source .venv/bin/activate
pip install -r backend/requirements.txt
pip install -r tests/backend-unit/requirements.txt
cd tests/backend-unit
pytest
‘‘‘

TROUBLESHOOTING

? **Port already in use**: change ‘GATEWAY_PORT‘ in ‘.env‘ (and restart: ‘docker compose down && docker compose up -d‘)
? **Containers unhealthy**: run ‘docker compose logs -f‘ and verify Postgres healthcheck passes
? **Database connection errors**: ensure ‘postgres‘ is healthy; check ‘PG_*‘ values in ‘.env‘
? **Uploads not accessible**: ensure ‘PUBLIC_MEDIA_URL‘ matches gateway URL and that ‘minio‘ + ‘minio-init‘ are running
? **AI features slow or failing**: leave ‘DEEPSEEK_API_KEY‘ empty to disable AI, or increase gateway timeout if enabling
? **Gmail not configured**: leave ‘GMAIL_*‘ blank; demo mode should still allow core flows to work